

SCIM — Signal Cognition Integrity Module

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Independent Methodological Authority

OriginID: OOF-OID-AI-SCIM-2026-06-03-0001

Architecture Ecosystem: Structured Reality Standards™

Architecture Family: Cognitive Governance Intelligence Architecture (CLIA®)

Operational Layer: Cognitive Interpretation Governance Layer

Governed Space: Signal Cognition Integrity

Category: AI & Interpretation

Subcategory: Signal Interpretation Governance

Type: Cognitive Interpretation Integrity Module

Parent Standard: Cognitive Interpretation Integrity Standard (CIIS)

Version: 1.0

Status: Canonical · Open Module

Origin Date: 3 June 2026

Compatibility: OOF Methodology OS ·

Cognitive Interpretation Integrity Standard (CIIS) ·

Cognitive Integrity Standard (CIS) ·

Multi-Layer Truth Validation Framework (MTVF) ·

Runtime Integrity Standard (RIS) ·

INTEGROS® — Integrity Standard

AI-Readable: Yes

Authority: OOF

Protection: MIP — Methodological Intellectual Property

Canonical Language: English (UCL)

Canonical Definition

Signal Cognition Integrity Module (SCIM) defines the structural conditions under which signals, observations, sensory inputs, events, data streams, environmental indicators, and operational reality inputs remain materially interpretable, traceable, contextually valid, and operationally reliable throughout the signal-to-meaning transformation process.

SCIM governs signal interpretation.

The module ensures that cognition remains capable of transforming signals into meaningful representations without introducing uncontrolled distortion, signal corruption, false significance assignment, or governance-invalid interpretation outcomes.

A system satisfies SCIM only if:

- signals remain identifiable
- signal origins remain traceable
- signal interpretation remains reconstructable
- signal relevance remains governable

- signal distortion remains detectable
- signal-derived meaning remains operationally valid

A cognition environment that cannot reliably interpret materially significant signals does not satisfy SCIM.

Module Operational Role

SCIM defines the signal-interpretation governance layer of CIIS.

Its role is to preserve valid signal understanding before higher-level interpretation, reasoning, planning, and decision-making occur.

Module Operational Space

SCIM governs:

- signal interpretation
- signal relevance
- signal traceability
- signal meaning formation
- sensory interpretation
- environmental signal interpretation
- event interpretation
- operational signal validity

The module applies wherever cognition receives information through signals requiring interpretation.

Module Function

The module applies wherever systems must preserve:

- signal validity
- signal understanding
- signal relevance
- traceable signal interpretation
- governance-valid signal processing
- operationally reliable meaning formation

Its function is to ensure that cognition interprets signals correctly before constructing meaning.

Minimum Implementation Framework

1. Define the Signal Object

The organization must define which signals require governance.

This may include:

- sensory signals
 - operational signals
 - environmental signals
 - event signals
 - machine-generated signals
 - human-generated signals
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- system alerts
- runtime observations

2. Define Signal Interpretation Conditions

The system must define the conditions under which signal interpretation remains valid.

This includes:

- signal relevance requirements
- signal traceability requirements
- interpretation requirements
- significance thresholds
- reliability requirements
- operational validity requirements

3. Define Signal Interpretation Degradation Detection Logic

The system must define how signal interpretation degradation is identified.

This may include:

- signal corruption
- signal ambiguity
- false signal prioritization
- signal-source uncertainty
- interpretation distortion
- relevance misclassification

4. Define Operational Response or Governance Logic

The system must define governance logic for signal-interpretation failures.

Governance response may include:

- signal review
- source verification
- interpretation reassessment
- escalation
- signal filtering
- governance intervention
- operational invalidation where required

5. Preserve Traceability & Restrict Invalid Conditions

The system must preserve reconstructable traceability of:

- signal sources
- signal classifications
- interpretation decisions
- governance reviews
- escalation actions
- resulting cognition outcomes

A cognition environment must not remain signal-valid if materially significant signals cannot be identified, interpreted, or traced.

Use Case 1 — Autonomous Monitoring Agent

Scenario

An autonomous monitoring agent continuously analyzes environmental inputs, operational alerts, runtime events, and infrastructure signals.

Application

SCIM governs signal interpretation validity, source traceability, and signal relevance throughout monitoring activities.

Result

The environment gains stronger signal reliability, reduced false interpretation risk, and improved operational awareness.

Use Case 2 — Autonomous Vehicle Perception Environment

Scenario

An autonomous vehicle continuously receives sensory information from cameras, sensors, environmental systems, and navigation inputs.

Application

SCIM governs signal interpretation, significance assignment, and signal-derived meaning formation before operational decisions occur.

Result

The environment gains stronger perception reliability, reduced signal distortion risk, and improved cognition validity.

Canonical Closing Statement

Signal Cognition Integrity Module (SCIM) defines the structural conditions under which signals, observations, sensory inputs, events, data streams, environmental indicators, and operational reality inputs remain materially interpretable, traceable, contextually valid, and operationally reliable throughout the signal-to-meaning transformation process.

Meaning cannot remain valid if signals are interpreted incorrectly.

Signal interpretation therefore becomes a foundational integrity condition of governable cognition.

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Canonical Source

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